Natalia Agapova

ML Engineer

natagapova.ru · Telegram · Email · Google Scholar · GitHub

Applied ML: NLP, explainability and debiasing, computer vision, RAG. I ship models with metrics, ablations, and honest trade-off analysis: accurate, explainable, and product-ready. Published research; I connect models to interfaces.

Experience

ML Software Engineer Apr 2025 – present
Croissan Studio
- AI products in a full-cycle studio: models, interfaces, and how results are presented.

Frontend Lead, ML Developer Feb – Sep 2024
Asimov Lab
- AI test-generation startup; front-end team lead. Full path from data to UI.

ML 2022 – present
Freelance
- Applied ML alongside design and web development.

Selected projects

Fair Resume Screening NLP · Debiasing · Paper
BERT · HeadHunter
- BERT resume classifier for 9 IT supercategories (HeadHunter). Integrated Gradients (Captum) for bias audit; 39+ debiasing configs. Baseline: 60.9% accuracy, city-swap flipped predictions in 7.7% of pairs.

Emotion Detection CV · FER2013 · On-device
EmotionCNN
- 58.7% accuracy, macro-F1 0.569; up to 60.8% with ensemble+TTA. INT8 quantization → Core ML and ONNX Runtime Web, 1.15 ms latency.

Gesture Input CV · MediaPipe · Real-time
Webcam Control
- Webcam cursor control via pinch / ready / click gestures; PyAutoGUI, smoothing, ~60 FPS.

Personal Knowledge System RAG · NLP · Python
Personal PDFs
- Answers from personal PDFs grounded in context only, with filename+page citations. Chunking, SentenceTransformer, ChromaDB, Ollama llama3.2.

Skills

Stack: Python, PyTorch · BERT, Captum · Debiasing, Counterfactuals · RAG, ChromaDB, Ollama · MediaPipe, OpenCV · Core ML, ONNX
Languages: Russian C2 (native) · English C1+ · French A2 · Korean A1

Education

Innopolis University 2026
BSc in Applied Artificial Intelligence

Publications

Improving Fairness in AI-Powered Recruitment: An Interpretable Resume Screening System
Agapova N. A.; Lukmanov R. A., 2026
doi:10.66693/mathai.1017

Quantum Annealing Approaches to Breaking RSA Encryption
Kholodov Y. A.; Salloum H.; Agapova N. A. Voprosy kiberbezopasnosti, 2025
doi:10.21681/2311-3456-2025-4-127-133